



**College of Computing and Informatics
Department of Computer Engineering**

Course Syllabus

Course Title

Introduction to Computer Eng.

Course Code
1502101

Fall 2026/2027

Course Syllabus

Course Title	Introduction to Computer Eng.				
Course Code	1502101				
Credit Hours	3	Contact Hours	4		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)					
Co-requisite(s)					
Semester	Fall 2026/2027				
Instructor Name	Email	Classroom	Section	Session Type	Time
Bilal Ahmed Arain	barain@sharjah.ac.ae	A8:101	31	Lecture Undergraduate	12:30 - 1345

Course Description:

This course covers an introduction to computer engineering as an integrated discipline among hardware, software, applications, and theory. Study of the basic architecture of computer systems includes information representation, computer hardware, software and programming languages, peripherals, storage systems, the concept of programming, high-level languages, e.g., C++, Python vs. low-level languages, e.g., assembly, operating system, embedded systems, computer networks and Internet, and computer security. The course includes introductory laboratory components, e.g., digital logic design lab, microprocessor lab, robotics lab, network lab, cloud computing lab, and computer programming. The course introduces the students to a perspective on the engineering profession, the engineering design process, problem-solving, and job prospects in computer engineering.

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Describe the concepts and structure of Computer Engineering labs and curricula.
2. Identify the components of a computer and the organization of those components
3. Analyze algorithms for problem solving in various computer programming languages
4. Identify key aspects of job prospects for computer engineers.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Describe ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	- Identify key aspects of job prospects for computer engineers.
Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	- Identify the components of a computer and the organization of those components - Analyze algorithms for problem solving in various computer

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
	programming languages
Apply new knowledge as needed, using appropriate learning strategies	- Describe the concepts and structure of Computer Engineering labs and curricula. - Identify the components of a computer and the organization of those components - Analyze algorithms for problem solving in various computer programming languages
Identify, formulate, and solve complex computer engineering problems by applying principles of engineering, science, and mathematics	- Describe the concepts and structure of Computer Engineering labs and curricula. - Identify the components of a computer and the organization of those components - Analyze algorithms for problem solving in various computer programming languages
Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	- Describe the concepts and structure of Computer Engineering labs and curricula. - Analyze algorithms for problem solving in various computer programming languages

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	What is the difference between Electrical Eng., Computer Eng., and Computer Science		1,4
2	-Computing and Technology Today (Computer History) -Hardware and Software Information Representation -Storage System		1
3	-Overview of CPE Courses and the Study Plan -Introduction to Digital Logic Design and the Number System		1
4	-Overview of CPE Courses and the Study Plan -Introduction to Digital Logic Design and the Number System		1,2
5	-Overview of CPE Courses and the Study Plan -Introduction to Digital Logic Design and the Number System		1,2
6	-Computer Architecture and the Little Man Computer -Introduction to Basic Assembly Language Programming		1,2,3
7	-Computer Architecture and the Little Man Computer -Introduction to Basic Assembly Language Programming		1,2,3
8	The CPU and Registers		1,2,3
9	The CPU and Registers		1,2,3
10	Operating Systems		1,2
11	C++ Programming Language Basics, Programming with C++ (Functions)		3
12	Programming with C++ (Control Structures)		3
13	High-level Programming using Python – Part A		3

Week	Topic	Comments*	CLOs
14	High-level Programming using Python – Part B		3
15	Final Exam		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Quizzes	TBA	25
Midterm Exam (No makeup Midterm. Students who miss the midterm with official excuse, their final exam grade will be calculated out of 75%)	TBA	30
Final Exam	TBA	45
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Describe the concepts and structure of Computer Engineering labs and curricula.	- Lectures - Demonstration - Laboratory Experiment	- Exams	Student average should be at least 70% for the CLO	- Innovation / creative thinking - Digital literacy - Numeracy	
Identify the components of a computer and the organization of those components	- Lectures - Demonstration - Laboratory Experiment	- Exams	Student average should be at least 70% for the CLO	- Innovation / creative thinking - Digital literacy - Numeracy	
Analyze algorithms for problem solving in various computer programming languages	- Lectures - Demonstration - Laboratory Experiment	- Exams	Student average should be at least 70% for the CLO	- Innovation / creative thinking - Digital literacy - Numeracy	
Identify key aspects of job prospects for computer engineers.	- Lectures	- Exams	Student average should be at least 70% for the CLO	- Innovation / creative thinking - Digital literacy - Numeracy	

Scope and Nature of Assessment Tools:

Quizzes: The quizzes help students consolidate their knowledge by revisiting and reviewing the topics covered in class. By engaging with quiz questions, students reinforce their understanding and solidify their knowledge of concepts covered in class.

Midterm and Final Exams: The exams help students consolidate their knowledge by revisiting and reviewing the topics covered in class.

Teaching and Learning Resources:

Textbook(s):

Warford J. S. (2017). Computer Systems. Jones & Bartlett Learning. 5th ed. -. ISBN:978-128407963.

Recommended Readings:

Fleddermann, C. B. and Bradshaw, M. D. (2003). Introduction to Electrical and Computer Engineering. 1st ed. Prentice Hall. ISBN:978-013033363.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).